Iroro **Orife**

Audio DSP, Speech · Research Engineer

(+1) 206-693-9979 | iroro@alumni.cmu.edu | ruohoruotsi | ruoho-ruotsi | iroro | Google Scholar

Objective

An applied research & dev role in Audio/Speech/NLP for Audio Post-Production or other frontier Audio/Speech ML applications

Skills

Languages	Python, C/C++, Java, bash, MATLAB, Objective-C, Scala, LaTeX. Proficient with OpenCode, Claude
DSP	Fourier Analysis, IIR/FIR Filter Design, MIR features, segmentation, audio source separation, ARM NEON SIMD
Speech	Training & maintenance of SAD, LID, Speech Intelligibility, LVCSR Kaldi & modern Transformer ASR models
Machine Learning	{D/C/R}NNs, Seq2Seq & multi-task modeling. Cross-modal CLIP/CLAP embeddings, Latent Diffusion, PCA/ICA/NMF
Audio	Extensive knowledge of audio engines, codecs, psychoacoustics, effects, mixing, film-audio production & post
Frameworks	PyTorch, TensorFlow, ONNX, Torchaudio, librosa, numpy, ffmpeg, CoreAudio, AudioUnit, Stagefright, Audioflinger
Infrastructure	Netflix {Spinnaker,Conductor}, Apache {Kafka, Spark}, MLOps, ModelHubs, Lambdas, MapReduce, Scaling in AWS EC2
Intangibles	I ship! Fast learner with strong engineering communication, team work ethic & emotional intelligence

Experience

Netflix — *Los Gatos, CA; Seattle, WA*

Senior Software Engineer - Audio Algorithms — *Nov. 2018 - Dec. 2026*

- Algorithm design & corpora construction for: Speech & Music Activity, Spoken Language & Dialect identification, \ Cinematic Audio Source Separation, Dialogue Intelligibility & AV Sync
- Trained, evaluated & productized models for inference in a Java/Python Microservice ecosystem
- For Subtitle/Dub authoring, computational evaluation of ASR & TTS models' real-world, customer-facing performance
- Conducted myriad Subjective Listening & Scaled AB Tests. Publications in ICASSP, ISMIR, CVPR, IEEE OJSP, EURASIP

Marchex — *Seattle, WA*

Staff Engineer — *Sept. 2014 - Nov. 2018*

- Applied research (from prototype to production) in spontaneous conversational telephone (Call Center) ASR
- Trained, deployed & maintained custom DNN-HMM models in Kaldi. Trained prod models for Language & VoIP-SPAM ID
- Performed 24/7 on-call support of Kaldi fleet processing 1M calls/day. Extensive experience with AWS EC2 {LB, ASG, EFS, EBS}

Dolby Laboratories — *San Francisco, CA*

Staff Audio Applications Engineer — *Jan. 2012 - Aug. 2014*

- Worked on embedded DSP implementations/ports of Dolby's perceptual audio coding & audio post-processing
- Algorithms: MDCT, spectral extension, hybrid complex QMF, dialogue, bass & volume leveling, surround virtualization
- Extensive experience conducting acoustic measurements & critical listening

Apple — *Cupertino, CA*

Senior Audio Software Engineer — *Feb. 2006 - Nov. 2011*

- **2006 - 2009**: Developed system level audio & effect processing features for Soundtrack Pro v2, Final Cut Studio 2 & 3
- Developed and maintained audio-unit hosting code for Logic Pro audio effects within ST Pro, Final Cut Pro X
- **2010-2011**: Developed & shipped an automatic track synchnoization feature & a multi-channel classifier for FCP X ingest
- Collaborated with the CoreAudio team on optimizations to their resampler & audio analysis algorithms

2KSports — *San Rafael, CA*

Audio Software Engineer — *May 2005 - Jan. 2006*

- Developed C/C++ tools & libraries for the XBOX360 launch of NBA2k6, NHL2k6, College Hoops 2K6
- Performed support tasks to game development teams using the audio library

Education

Dartmouth College — *Hanover, NH*

A.M. in Music Composition (Electro-Acoustic Music) — *Sept. 1999 - June 2001*

- Interdisciplinary Masters in 20th century Composition, Music Cognition, Aesthetics, Acoustics & Signal Processing

Carnegie Mellon University — *Pittsburgh, PA*

B.Sc. in Electrical and Computer Engineering, with a double major in Mathematics/Computer Science — *Sept. 1993 - May 1997*

École Polytechnique Fédérale de Lausanne (EPFL) *Lausanne, Switzerland*

3rd Year Exchange Student in Computer and Communication Sciences *July 1995 - August 1996*

Selected Publications

2024 **DJ Tool Retrieval Using Speech Activity, Music Structure And CLAP Embeddings**, ISMIR LBD 2024
2024 **Leveraging Counterfactual Cross-modal Pairs For Audiovisual Representation Learning**, CVPR 2024
2024 **Ìròyìnspeech: A Multi-purpose Yorùbá Speech Corpus**, LREC-COLING 2024
2023 **A Generalized Bandsplit Neural Network For Cinematic Audio Source Separation**, ICASSP-OJSP 2023
2022 **BibleTTS: A Large, High-Fidelity, Multilingual, And Uniquely African Speech Corpus**, Interspeech 2022
2022 **A Large TV Dataset For Speech And Music Activity Detection**, EURASIP-JASMP 2022
2020 **Masakhane – Machine Translation For Africa**, AfricanNLP Workshop (ICLR 2020)
2018 **Attentive Seq2Seq Learning For Diacritic Restoration Of Yorùbá Text**, Interspeech 2018
2018 **The Marchex 2018 English Conversational Telephone Speech Recognition System**, arXiv
2017 **Semi-Supervised Model Training For Unbounded Conversational Speech Recognition**, arXiv
2017 **Audio Spectrogram Factorization For Classification Of Telephony Signals**, arXiv
2001 **Riddim: Rhythm Analysis Based On Independent Subspace Analysis**, Dartmouth College A.M. Thesis

Patents

2023 **US 18/505,081**, Methods and systems for learning language-invariant audiovisual representations *Netflix*
2020 **US 10,810,995 B2**, Automatic speech recognition (ASR) model training *Marchex*
2019 **US 9,485,354**, Identifying call features and associations to detect call traffic SPAM *Marchex*
2015 **US 8,929,561**, System and Method for Automated Audio Mix Equalization *Apple*
2014 **US 8,842,842**, Detection of audio channel configuration *Apple*
2013 **US 8,621,355**, Automatic synchronization of media clips *Apple*

Extracurricular

IEEE, AES, ISCA

Member of Professional Electrical & Audio Engineering Societies *2012 - PRESENT*

Niger-Volta Language Technologies Institute

Founder of open-source, West African, Language technology organization *2018 - PRESENT*

Independent Record Label(s)

Founder, Record Producer, Recording & Mix Engineer, Label Manager, A&R, Staff writer & DJ *2003-2012, 2019 - PRESENT*

Pacific Northwest Film Scoring (Evening Program)

Part-Time Student *September 2026 - PRESENT*

Natural Languages

French

Full Professional Proficiency, written & spoken

- University of Lausanne coursework in Literature & Film
- ILR proficiency of 4+

Yorùbá

Conversational Proficiency

- Junior secondary school reading/writing level

Spanish

Basic Proficiency

- Fulfill traveling needs & conduct myself in a polite manner